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MILITARY SPECIFICATION

WATER SOFTENER UNIT, ZEOLITE PRESSURE TYPE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers general purpose sodium cycle and hydrogen cycle zeolite pressure type water softening equipment. The design and size shall be as specified (see 6.2).

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

GG-G-76 - Gages, Pressure and Vacuum, Dial Indicating, (For Air, Steam, Oil, Water, Ammonia and Chloro-Fluoro Hydrocarbon Gases).
QQ-S-781 - Strapping, Steel, and Seals.
TT-E-527 - Enamel, Alkyd, Lustreless.
TT-V-121 - Varnish, Spar, Water-Resisting.
UU-T-81 - Tags, Shipping and Stock.
PPP-B-35 - Bags, Textile, Shipping, Burlap, Cotton and Waterproof Laminated.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock Corner.
PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-C-843 - Cushioning Material, Cellulosic.
PPP-P-40 - Packaging and Packing of Hand Tools.

MILITARY

MIL-V-3 - Valve, Fittings, and Flanges (Except for Systems Indicated Herein), Packaging of.
MIL-C-104 - Crates, Wood; Lumber and Plywood Sheathed, Nailed and Bolted.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4610

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MIL-P-116 - Preservation-Packaging, Methods of.
MIL-B-121 - Barrier Material, Greaseproofed, Waterproofed Flexible.
MIL-C-3774 - Crates, Wood; Open 12,000- and 16,000-Pound Capacity.
MIL-E-16298 - Electric Machines Having Rotating Parts and Associated Repair Parts: Packaging of.
MIL-E-17555 - Electronic and Electrical Equipment Accessories, and Repair Parts, Packaging and Packing of.
MIL-T-22085 - Tapes, Pressure-Sensitive Adhesive, Preservation and Sealing.
MIL-C-52950 - Crates, Wood, Open and Covered.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-147 - Palletized Unit Loads.
MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.
MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of.
MIL-STD-1186 - Cushioning, Anchoring, Bracing, Blocking and Waterproofing, with Appropriate Test Methods.

(Copies of specification, standards, drawings, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D1126 - Hardness in Water, Methods of Test for.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Boiler and Pressure Vessel Code
Section VIII - Pressure Vessel - Division 1
- Pressure Vessels Alternate Rules - Division 2

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017)

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AMERICAN WATER WORKS ASSOCIATION (AWWA)

C700 - Standard for Cold Water Meters - Displacement Type
C701 - Standard for Cold Water Meters - Turbine Type for Customer Service

(Application for copies should be addressed to the American Water Works Association, 2 Park Avenue, New York, NY 10016.)

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

MG-1 - Motors and Generators

(Application for copies should be addressed to the National Electrical Manufacturers Association, 2101 L Street, NW, Washington, DC 20037.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, N.W., Washington, DC 20036.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Tariff Publishing Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

3. REQUIREMENTS

3.1 Description. The softener unit shall be of the downflow pressure type with automatic or manual controls as specified (see 6.2). The unit shall operate on the sodium or hydrogen cycle, as required, by the condition of the raw water and in consideration of the intended use. The water softener unit shall be comprised of a softener shell, distribution system, resin, an underdrain system, regenerant tanks, regenerant measuring system, valve system, and all controls, fittings, and interconnecting piping to make complete integrated units designed for operation of water softening.

3.2 First article. When specified (see 6.2), the contractor shall furnish a water softener equipment for first article inspection and approval (see 4.3 and 6.6).

3.3 Standard commercial product. The water softener unit shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product. Additional or better

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features which are not specifically prohibited by this specification but which are a part of the manufacturer's standard commercial product, shall be included in the unit being furnished. A standard commercial product is a product which has been sold or is being currently offered for sale on the commercial market through advertisements or manufacturer's catalogs, or brochures, and represents the latest production model.

3.4 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification unless otherwise specified.

3.5 Softener tank. The softener shell shall be steel cylinders with dished heads. All seams shall be welded. The tank shall conform to Section VIII of the ASME Boiler and Pressure Vessel Code. Unless otherwise specified (see 6.2), the tank shall be designed for a working pressure of 100 pounds per square inch (psi) and tested to withstand a hydrostatic pressure of 50 percent in excess of the design working pressure. Tanks 30-inches in diameter or less shall be equipped with two handholes, with size no less than 4-inch by 6-inch, one to be located in the top head, and the other to be located in the side-sheet near the bottom of the tank. Tanks larger than 30-inches in diameter shall have a manhole no less than 11-inch by 15-inch in the top head of the tank, and a handhole no smaller than 4-inch by 6-inch near the bottom of the sidesheet or when specified (see 6.2), shall have a coupling on bottom head in lieu of 4 by 6 handhold. All tanks shall have adjustable jack-type legs, or structural approved steel legs.

3.5.1 Linings. Unless otherwise specified (see 6.2), the tanks operating on the sodium cycle shall not be lined. If required, the lining shall be of plastic material having an overall thickness not less than 0.005 inch. Tanks operating on the hydrogen cycle shall be rubber-lined or plastic (polyvinyl-chloride (PVC)) lined. The rubber lining shall be not less than 0.1875 inch thick and the plastic liner shall be not less than 0.06 inch thick.

3.6 Exchange resin. The cation exchange resin shall be a high capacity styrene vinyl-benzene containing sulfonic acid groups with an effective size of 0.45-0.60 millimeters and a uniformly coefficient not greater than 1.8. The exchange capacity range for the sodium cycle shall be 18-32 kilograms per cubic foot when regenerated with from 5.5 to 15 pounds of salt. The exchange capacity for the hydrogen cycle shall be not less than 11 kilograins pounds per cubic foot (pcf) using 5 pounds of sulfuric acid pcf applied as a 2 percent solution. The exchange capacity shall be not less than 9-13 kilograins pcf when use is made of 4.5 pounds of 15 percent hydrochloric acid. The above exchange capacities are indicated for purposes of determining the required amount of resin to be installed in the unit. The minimum bed depth acceptable is 24 inches.

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3.7 Distribution system. Each shell shall have a top inlet distributor consisting of the header-lateral type or baffle plate designed to evenly distribute the incoming raw water over the entire zeolite bed, and to uniformly collect the waste water coming up through the zeolite during the washing operation. This inlet may also serve as the backwash outlet and should be at a distance above the bed equal to 50 percent of the depth of the resin. The inlet for the sodium cycle may be wrought iron, brass pipe containing not less than 85 percent copper or suitable plastic. Hydrogen cycle units shall have corrosion-resistant steel pipe, rubber coated steel, plastic-coated steel, or plastic pipe such as type 11 PVC.

3.8 Underdrain systems. Each softener shell shall be provided with an underdrain system for the collection of softened water. The underdrain system in the bottom of each softener unit shall be of the header-lateral type with iron-pipe-size red brass type laterals or of the header-strainer nozzle type with brass or stainless steel nozzles or of PVC hubs type lateral. While in operations the system shall not become clogged with dirt and sand. The system shall distribute the backwash water laterally in a way that will not cause channeling of the sand and gravel bed, if used. For use in the hydrogen cycle the system shall be made of rubber-covered steel, corrosion-resistant steel, hard rubber, or plastic covered steel.

3.9 Main operating valve. The transfer of both water and regenerant to and from the zeolite pressure tank shall be accomplished by a valve system consisting of a single multiported valve using a single rotary disk or multiple reciprocating disks - or by a nest of individual valves using a single reciprocating disk for each valve, pre-piped into a single detachable unit. The valve shall be of proper size to affect the regeneration of the zeolite resin. The service flow rate shall be controlled with regard to flow in the "SERVICE" cycle or no-flow in any of the regeneration cycles by the main regeneration valve-pilot assembly and the service flow may be through the main regeneration valve-pilot assembly or through reciprocating, single disk, type service valves when the service flow rate is higher than the capability of the multiport valve.

3.9.1 Multiport valve. The multiport valve shall contain piping connections for the raw water inlet, treated water outlet, (unless hi-flow service valves are required), transfer to the top of the zeolite tank, transfer to the bottom of the zeolite tank, regenerant inlet, and waste (drain) outlet. The valve shall have internal provisions to control the proper direction of flow, and rate of flow, in such a manner as to avoid sudden inrushes of water or regenerant which will cause damaging shocks to the equipment. Valves designed for manual use shall be operable by means of a single handle and the torque required to index this handle to the various valve positions shall not be excessive. The design shall permit selective operation of any phase of the regeneration cycle. Valves designed for automatic operation shall be controlled by a hydraulic, pneumatic, electric, or combination pilot device which will index the multiport valve to the proper positions to accomplish the regeneration cycles. The design shall permit the various steps of operation service, backwash, regenerant flow, and rinse, to be accomplished by the rotation of a shaft which causes the opening and closing of parts in correct sequence. The automatic part sequence control shall be accomplished by meter contactor or push-button or other impulse-initiating device. Manual operation of the valve shall be accomplished either

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during the normal operation of the controller or during loss of electrical power by means of a hand operated pilot by-pass assembly. Hand operated valves shall be installed before the inlet, and after the outlet of the multiport valve (or hi-flow service valves if required) to stop the flow of water through the softener unit in order to perform maintenance as required. Unions or flanges shall be installed to accomplish removal of the control valves, when necessary. All parts of the multiport valve including trim, exposed to influent water or regenerant flow shall be of corrosion-resistant to the solutions handled. The use of acid or caustic soda regenerant solutions require that metals shall not be of such divergence in the metallic series as to cause galvanic action. Monel metal shall not be required in the caustic soda regenerated cycle.

3.9.2 Package valve system. The package of valves for each zeolite tank shall consist of a compact nest of individual valves with fittings, mounted in such a manner as to make a single complete unit. Systems designed for manual use only shall consist of individual hand operated valves, pneumatic, or hydraulic operated individual valves operated by a single hand operated pilot device. Systems designed for automatic use shall consist of individual pneumatically or hydraulically operated valves controlled by a single electrically driven pilot. Provision shall be made such that the automatic pilot device can be controlled manually by means of a hand operated dial or wheel attached to the pilot during normal operation of the softener or during the loss of electrical power. A hand operated valve shall be placed before the raw water inlet valve and after the treated water outlet valve to stop the flow of water through the softener unit in order that normal maintenance can be performed. Unions or flanges shall be installed into the valve system to allow removal of the valve nest as necessary. The control shall permit selective operation of any phase of the cycle and shall lend itself to automatic operation from line water meter, remote pushbutton or other impulse-initiating device. The treatment and metals used for the package type valve shall be as specified for the multiport valve. Unless otherwise specified (see 6.2), the valve shall be designed for a working pressure of 100 psi and shall withstand a test pressure 1.5 times the working pressure.

3.9.3 Automatic operation. When automatic operation is specified, all additional mechanical and electrical equipment required shall be supplied. The valve shall be turned successively to the backwash, regenerate rinse, and service position, and held for the correct length of time. The supplementary operations consisting of backwashing, feeding of regenerant, and rinsing, shall be started automatically as the valve is turned to the appropriate position. Multiple unit installations shall be provided with an interlocking system which prevents regeneration more than one unit at a time. In case of power failure the units shall be operable manually, and the controls shall be adjustable.

3.10 Regeneration equipment. Each softener shall be furnished with a combination salt saturator-measuring tank. The tanks shall be constructed of corrosion free, rigid polymer, fiberglass construction, or steel. Steel tank shall be either hot-dipped galvanized inside and outside or suitably protected against corrosion with lining and coating material.

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3.10.1 Brine regenerating system. A single steel combination saturator-measuring tank with gravel filter, collecting system, transfer ejector, and necessary valving, and a bronze float gage for indicating the correct amount of salt brine shall be provided with each installation. A layer of gravel or anthracite shall be provided with the steel tank painted with rust-inhibiting primer. The tank shall hold enough salt for at least eight regenerations. The necessary controls for maintaining the liquid level in the tank, together with wrought iron pipe, brass pipe containing not less than 85 percent copper, or type II PVC pipe, shall be supplied for connection of valves, tank and softener.

3.10.2 Brine distributors. Brine distributors, if necessary for proper operation of softener, shall be header and lateral arrangements or hub lateral constructed of PVC material or of wrought iron pipe or brass pipe containing a minimum of 85 percent copper. The outlet nozzles shall be not more than 6 inches above the upper surface of the zeolite bed.

3.10.3 Ejector. When specified (see 6.2), an ejector operated by water supplied under pressure, shall draw the brine into the softener, diluting the brine to the proper concentration. The associated valves necessary for the efficient operation shall be supplied. The ejector and valves shall be bronze or brass containing not less than 85 percent copper, corrosion-resistant polymer, or PVC material.

3.10.4 Acid regeneration equipment. A measuring tank and a dilution tank shall be provided with each installation for dilution of the acid to the proper concentration. The tanks shall be lead-lined, rubber-lined, polyethylene material, or corrosion-resistant steel. The measuring tank shall be large enough to hold a charge for one regeneration. All necessary pipe, valves, and ejector operated by water pressure (see 6.2) to introduce the acid solutions into the hydrogen shall be furnished.

3.10.5 Acid distributors. Acid distributors, shall be header and lateral arrangements constructed of corrosion-resistant steel, hard rubber, or plastic pipe, or lined and coated with hard rubber or plastic. The outlet nozzles shall be not more than 6 inches above the upper surface of the zeolite bed.

3.10.6 Acid transfer system. When specified (see 6.2), a complete system for transferring acid from carboys or drums to the acid measuring tank shall be provided. If fluid pressure is used to transfer the acid from the carboys or drums, it shall not exceed 1 psi and a safety valve set at 1 psi shall be included in the equipment.

3.11 Proportioning equipment. When the condition required of the softened water is such that all the raw water need not be softened, proportioning equipment shall be supplied. The equipment, shall be fully automatic and shall maintain the correct proportion with fluctuating line pressures. The system shall be either pneumatically operated or hydraulically operated and controlled. The equipment shall be adjustable to allow for changing water conditions. The equipment shall include, on the sodium cycle, a low-flow contactor which will stop the total flow if for any reason the flow from the sodium cycle should stop.

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3.12 Zeolite supporting beds. The supporting beds shall be either gravel or anthracite.

3.12.1 Gravel beds. Gravel supporting beds shall be clean graded quartz. The beds shall be provided with normal standard sizing of gravel for proper operation of the underdrain system. Gravel beds shall not be used in softeners operating the hydrogen cycle except when the silica pickup is not critical.

3.12.2 Anthracite beds. Anthracite supporting beds shall be clean graded anthracite. The anthracite supporting beds shall be made up of normal standard sizing of anthracite for proper operation of the underdrain system. Anthracite beds shall be used in all hydrogen cycle softeners.

3.13 Electric motors. Unless otherwise specified (see 6.2), the motor, excluding timers and fractional horsepower motors shall be 220-volt, 3-phase, 60-cycle, totally enclosed, class A insulation motors, equipped with thermal overload elements. The motors shall be the motor manufacturer's recommended size for the power requirements. The temperature rise shall not exceed 55° Celsius (C) when the motor operates at full load in an ambient-temperature of 40°C. The motors shall conform to NEMA MG-1.

3.14 Water meter. Each softener unit shall be provided with a water meter. A displacement type water meter shall be required for inlet pipes under 2-inches, and over 2 inches a current type shall be used. The rated capacity of the water meter shall be 20 percent greater than the rated capacity of the unit. The displacement type meter shall conform to AWWA C700. The current type meter shall conform to AWWA C701.

3.15 Degasifiers. A degasifier of the trickle type shall be supplied for hydrogen cycle softeners. When specified (see 6.2), the degasifier shall use counterflow of water and air and shall reduce the CO_2 content of the water to less than 10 parts per million. The degasifier shell shall be constructed of wood staves and filled with wooden slats or trays arranged in staggered formation. The water shall be discharged into the top of the shell through nonclogging corrosion-resistant steel, fiberglass, or plastic nozzles. A fan and motor adequate to supply sufficient air for the service required shall be supplied. All interconnecting valves and piping shall be included.

3.16 Testing equipment. Each unit shall be furnished with a complete testing outfit adequate for testing of total hardness and pH. The range for pH testing shall be from 4 to 10.

3.17 Length of cycle. Unless otherwise specified (see 6.2), softeners shall be designed for regeneration at a frequency not exceeding one every 24 hours when manually operated, or exceeding once every 8 hours when operated automatically.

3.18 Regeneration.

3.18.1 Sodium cycle. The sodium cycle softeners shall be regenerated with a salt solution (NaCl). The salt requirements shall not exceed the following figures at the exchange capacities specified in section 3.6. Styrene base zeolites -0.3 to 0.55 pound salt per kilograin (1,000 grains) of hardness exchanged.

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3.18.2 Hydrogen cycle. The hydrogen cycle softeners shall be regenerated with a solution of sulfuric acid or hydrochloric acid. The acid requirements shall not exceed the figures at the exchange capacities specified in section 3.6. The operation of the hydrogen cycle unit shall be such that calcium sulfate precipitation does not occur.

3.18.3 Regeneration time. The capacity of the regeneration equipment shall be large enough to completely regenerate a sodium cycle softener in less than 1.0 hour and a hydrogen cycle softener in less than 1.5 hours when depth of resin bed is less than 60 inches.

3.19 Pressure loss. The pressure loss through sodium cycle and hydrogen cycle softeners measured from ahead of the raw water meter to be softened water outlet shall not exceed 35 feet of water. Two pressure gages conforming to GG-G-76 shall be mounted on the equipment, one in the raw water line and one in the soft water outlet.

3.20 Performance. Unless otherwise specified (see 6.2), the hardness of the effluent shall be reduced to zero as measured by the standard Boutron and Boudet soap test or in accordance with the manufacturer's standard levels of measurement as specified in units of $CaCo_3$ when the equipment is operating at rated capacity with a raw water approximately of the analysis specified in the procurement documents. The color and turbidity of the effluent from the softener shall in no case be greater than the raw water. Slugs of dirty or turbid water shall not be delivered at any time by the equipment from zero flow to the rated capacity. Tests to determine conformance with the requirements shall follow the procedures specified in ASTM D1126.

3.21 Electromagnetic interference control. When specified (see 6.2), the equipment shall conform to the electromagnetic interference control requirements and test limits for class No. IIIC equipment as specified in MIL-STD-461.

3.22 Treatment and painting. Unless otherwise specified (see 6.2), the equipment shall be treated and painted in accordance with the manufacturer's best standard practice.

3.23 Identification marking. Equipment, assemblies, parts, and manufacturer, installer, and contractor's name and address shall be marked for identification in accordance with the manufacturer's standard practice.

3.24 Instruction plates. Each unit of equipment shall be equipped with instruction plates, suitably located, describing any special or important procedure to be followed in operating and servicing the equipment. Plates shall be of copper or zinc-base-alloy, adequately secured to the unit.

3.25 Lifting attachments and tiedown devices. When specified (see 6.2), suitable lifting attachments and tiedown devices shall be furnished in accordance with the manufacturer's standard practice except as specified herein. The attachments and devices shall have eyes not less than 3 inches in inside diameter, and plates to withstand any handling conditions encountered. Adequate tiedown devices may also be used for lifting purposes. Each attachment of tiedown device shall be of sufficient capacity to carry the total weight. A copper or zinc-base-alloy, or aluminum plate, securely fastened to the unit, shall indicated diagrammatically the lifting capacity of each attachment.

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3.26 Drawings and technical publications. When specified (see 6.2), the manufacturer shall submit installation drawings and shall furnish operating instruction manuals with lubrication chart, parts book, and repair manuals applicable to the equipment and the attachment with each unit in reference to section 6.5 of this specification.

3.27 Repair or maintenance parts, maintenance tools, and accessories. When specified (see 6.2), repair or maintenance parts, maintenance tools, and accessories shall be furnished, applicable to the prime equipment.

3.28 Workmanship. The quality of workmanship shall be such as to produce softeners that are in accordance with the requirements of this specification and insure proper functioning of all parts of the unit.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Govermenmt. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. Inspection shall be classified as follows:

(a) First article inspection (see 4.3)
(b) Quality conformance inspection (see 4.4)
(c) Packaging inspection (see 4.7)

4.3 First article inspection. First article inspection shall be performed on one softening unit when a first article sample is required (see 3.2 and 6.2). This inspection shall include the examination of 4.5 and the tests of 4.6. The first article may be a first production item or a standard production item from the contractor's current inventory provided the unit meets construction, and manufacturing technique applicable to the remaining unit to be furnished under the contract.

4.4 Quality conformance inspection. Quality conformance inspection shall include the examination of 4.5 and the tests of 4.6.

4.5 Examination. Each water softener unit shall be examined for compliance with the requirements specified in Section 3 of this specification. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examination and dimensional measurements. Noncompliance with any specified requirements or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

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4.6 Tests.

4.6.1 Shell test. After the units are installed, the shell shall be tested in accordance with the ASME Boiler Pressure Vessel Code for the Construction of Unfired Pressure Vessels.

4.6.2 Performance test. The unit in an installation, after being put into operation, shall be tested to determine conformance with this specification. With a raw water analysis of approximately that specified in the procurement documents, the effluent analysis shall fall within the limits specified in the procurement documents.

4.6.3 Operating tests. Before the test is started, each softener shall be run to exhaustion and then regenerated to full capacity in accordance with manufacturer's instructions. The softener shall be put through two complete cycles at the actual demand rate for working test. During the capacity tests, the softened water will be sent to the sewer, if necessary, to maintain the required flow rate. After each run, the unit shall be regenerated with use of salt brine delivered from the measuring tank in the amount called for by the operating instructions. At the end of the brine rinse and beginning of production of softened water, a 1/2-pint sample of softened water shall be taken every 15 minutes and used to make up a complete sample. From each test run composite sample, a 1 quart sample shall be taken which shall be tested, and the result of the test shall fall within the limits specified in the procurement documents.

4.6.3.1 Color and turbidity. Under actual operating conditions, the exchange material shall not be washed out of the apparatus and turbidity, and color of the softened water shall not exceed the turbidity and color of the hard water. During any softening run, slugs of dirty or turbid water shall not be delivered regardless of the change of demand rate up to the maximum on the apparatus. During a run, the sampling petcock shall remain open, allowing a stream of treated water to be collected in a wide-mouth bottle set against a white background, so that the color and turbidity may be under observation at all times. If the turbidity and color of the treated water exceeds the turbidity and color of the untreated water, the Government reserves the right to reject the equipment in its entirety and required installation of other equipment that will meet the specification requirements.

4.6.4 Electromagnetic interference control test. The preproduction model, equipped for the reduction of radio interference in accordance with 3.20 shall be tested by the contractor in accordance with Test Method RE05 of MIL-STD-462. The Government reserves the right to witness tests performed by the contractor or his private testing organization. The contractor shall furnish the contracting officer, for approval, the Interference Control Plan, the EMI/EMC Test Plan and the Electromagetic Test Report required in MIL-STD-461. Upon approval of the Test Report by the contracting officer and provided all other requirements of the specification are met, the preproduction sample or initial unit of production, as applicable, shall be used as a model for all production units.

4.7 Packaging inspection. The preservation, packaging, packing and marking of the equipment shall be inspected to determine conformance with the applicable requirement of Section 5 of this specification. The preservation,

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packaging, packing, and marking requirements of section 5 identified to referenced specifications and standards shall be inspected in accordance with the applicable requirements of these documents. The preservation and packaging requirements of Section 5 not identified to referenced specifications shall be examined for the defects listed in 4.7.1.1 on the basis of the sampling procedures and inspection levels being used to determine examination requirement under MIL-P-116. The packing and marking requirements of Section 5 not identified to referenced specifications shall be examined for the defects listed in 4.7.1.2 and 4.7.1.3 in accordance with MIL-STD-105 based on an Acceptable Quality Level of 4.0 percent defective.

4.7.1 Classification of defects.

4.7.1.1 Preservation and packaging.

Major

101 When removed for disassembly, bolts, nuts, screw, and washers not reinserted in one of the applicable mating parts.
102 Parts removed and mating parts not matchmarked as specified.
103 Openings not protected as specified.
104 Closure of fiberboard containers incorrect.
105 Fiberboard box not waterproof sealed.
106 Cushioning material missing when required.
107 Gross weight of fiberboard container exceeds limit specified.

4.7.1.2 Packing

Major

108 Shrouds missing.
109 Closure of shipping container incorrect.
110 Strapping missing or incorrectly applied.
111 Strapping not specified type and class.
112 Cushioning, blocking, and bracing not as specified.

4.7.1.3 Marking.

Major

113 Marking missing, illegible, or incorrect.

5. PACKAGING

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Disassembly. In order to reduce the size of the shipping container or to provide additional protection, projecting parts which can be easily reassembled shall be removed. All nuts, bolts, pins, and washers involved shall be replaced in one of the mating parts and secured to prevent loss.

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5.1.1.2. Matchmarking. To facilitate reassembly, parts removed and the mating parts on the basic unit shall be matchmarked with lustreless white enamel conforming to TT-E-527. Matchmarking shall be overcoated with clear spar varnish or clear lacquer conforming to TT-V-121. Small parts that are removed shall have matchmarking information on cloth shipping tags conforming to UU-T-81, type A, attached to mating parts. The marked tags shall be waterproofed in accordance with MIL-STD-129.

5.1.1.3 Preservative application. All unpainted critical, operating, or highly finished ferrous metal sufaces shall be coated with P-9 preservative oil in accordance with MIL-P-116. All other unpainted ferrous metal surfaces including threaded areas shall be coated with P-1 corrosion preventive in accordance with MIL-P-116. All threaded openings shall be sealed with threaded metal or plastic caps or plugs and all flanged openings shall be sealed with blank flanges, using greaseproof barrier material conforming to MIL-B-121 between the metal surfaces. All other openings shall be sealed with type I, grade C greaseproof barrier material conforming to MIL-B-121, secured in place with type 1 tape conforming to MIL-T-22085.

5.1.1.4 Valves, fittings, and pipe. Valves, fittings, and pipe shall be preserved and packaged in accordance with the applicable level A requirements of MIL-V-3.

5.1.1.5. Meters and gages. The meters and gages shall be preserved by method IC-2 in accordance with MIL-P-116.

5.1.1.6 Electrical equipment. Electrical equipment shall be preserved and packaged in accordance with the applicable level A requirements of MIL-E-16298 and MIL-E-17555.

5.1.1.7 Pump. All interior surfaces of the housing and all parts or assemblies within the housing shall be coated with P-14 preservative in accordance with MIL-P-116 applied by spraying or fogging through the suction and discharge openings while the pump is actuated by hand. Drain excess preservative from the housing. Coat the drain plug with P-1 preservative in accordance with MIL-P-116, and replace.

5.1.1.8 Exchange material and supporting bed material. Exchange material and supporting bed material shall be packaged in bags conforming to PPP-B-35, type II.

5.1.1.9 Testing apparatus. Testing apparatus shall be packaged in fiberboard boxes conforming to PPP-B-636 (weather-resistant class) and cushioned with cellulosic cushioning material conforming to PPP-C-843, type 1, class A. The boxes shall be waterproofed sealed with tape in accordance with the appendix to the box specification.

5.1.1.10 Tools. Tools shall be preserved and unit packaged in accordance with the applicable level A requirements of PPP-P-40.

5.1.1.11 Technical publications. Technical publications shall be preserved by method IC-1 in accordance with MIL-P-116.

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5.1.1.12 Repair parts and accessories. Repair parts and accessories shall be preserved and packaged in accordance with the applicable level A requirements of MIL-V-3, MIL-P-16298, and MIL-E-17555 and with an applicable submethod of MIL-P-116 for those parts and accesssories not specifically covered by the aforementioned specifications.

5.1.1.13 Level C. The complete equipment shall be preserved and packaged in accordance with supplier's standard practice.

5.2. Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Shells and tanks. Shells and tanks shall be packed in crates conforming to MIL-C-104, type I, style a. Contents shall be cushioned, blocked, braced, and anchored in accordance with MIL-STD-1186 and MIL-C-104.

5.2.1.2 Valves, fittings, pipe, meters, gages, electrical equipment, pumps, testing apparatus, tools, and technical publications. All valves, fittings, pipe, meters, gages, electrical equipment, pumps, testing apparatus, tools, and technical publications shall be packed in cleated plywood boxes conforming to PPP-B-601, overseas type, or nailed wood boxes conforming PPP-B-621, class 2. Contents shall be cushioned, blocked, and braced in accordance with MIL-STD-1186. Flat steel strapping shall be type I, class B conforming to QQ-S-781.

5.2.1.3. Palletization. Bags of anthracite shall be palletized in accordance with MIL-STD-147, load type XV.

5.2.2 Level B

5.2.2.1 Shells and tanks. Shells and tanks shall be packed in open wood crates conforming to MIL-C-52950 or MIL-C-3774. Contents shall be shrouded in accordance with the applicable crate specification. Contents shall be cushioned, blocked, and braced in accordance with MIL-STD-1186.

5.2.2.2 Valves, fittings, pipe, meters, gages, electrical equipment, pumps, testing apparatus, tools, and technical publications. All valves, fittings, pipe, meters, gages, electrical equipment, pumps, testing apparatus, tools, and technical publications shall be packed in cleated plywood boxes conforming to PPP-B-601, domestic type, or nailed wood boxes conforming PPP-B-621, class 1. Contents shall be cushioned, blocked, and braced in accordance with MIL-STD-1186.

5.2.2.3 Palletization. Bags of anthracite shall be palletized in accordance with MIL-STD-147, load type XV.

5.2.3 Level C. The complete equipment shall be packed in a manner which will insure arrival at destination in satisfactory condition and be acceptable to the carrier at lowest rates. Containers and packing shall comply with Uniform Freight Classification rules or National Motor Freight Classification rules.

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5.3 Marking. In addition to any special marking required, interior packages and shipping containers shall be marked in accordance MIL-STD-129.

6. NOTES

6.1 Intended use. Zeolite water softeners are general purpose softeners suitable for a wide variety of installations, both large and small. Water is conditioned by running it through an exchange material. Exchange materials are commonly regenerated with either salt or acid, and selection of the proper type is dependent on the condition of the water. Units regenerated with salt are termed sodium cycle units. Those regenerated with acid are termed hydrogen cycle units. Softeners operating on the sodium cycle reduce, hardness. Those operating on the hydrogen cycle reduce hardness, alkalinity and total dissolved solids. The two cycles can be operated independently or in parallel. Unless otherwise directed, the manufacturers shall choose the combination necessary to produce water of the specified quality from the raw water of the specified quality.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Design information (see 1.1):

(1) Service in which softener will be used.
(2) Source of raw water (name of river, name of city water supply, geographic location of well, etc.)
(3) Raw water analysis- give quantities of the following substances subtances in parts per million or grains per gallon: calcium, magnesium, sodium, silica, iron oxide and alumbina, bicarbonate, carbonate, hyydroxide, sulphate, chloride, phosphate, volatile and organic matter, total solids in solution, suspended matter, CO_2 - free, hardness in terms of $CaCo_3$, free acid, pH, color.
(4) Effluent analysis; specify the allowable solids in solution, hardness in terms of $CaCo_3$, and pH required of the softened water (see 3.25).
(5) Design water flow.
(6) Maximum water temperature.

(c) Whether automatic or manual controls are required (see 3.1).
(d) When first article is required for inspection and approval (see 3.2, 4.3, and 6.6).
(e) What shell and valve design pressure should be if other than 100 psi (see 3.5 and 3.9.2).
(f) When tank is to have a coupling on bottom head in lieu of 4 by 6 handholes (see 3.5).
(g) Whether sodium cycle softener shells shall be lined (see 3.5.1).
(h) Water pressure available for operating ejectors (see 3.10.3 and 3.10.4).
(i) Whether acid transfer system is required; if so, what location (see 3.10.6).
(j) Type of motor, if other than that specified (see 3.13).
(k) Type of basin material (see 3.15).
(l) Length of cycle, if other than that specified (see 3.17).

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(m) Whether hardness of effluent shall be reduced to zero (see 3.20).
(n) Whether radio-interference suppression is required (see 3.21).
(o) Color if other than that specified (see 3.22).
(p) Lifting attachment, if necessary (see 3.25).
(q) When drawings and technical publications are required (see 3.26)
(r) Applicable lists of repair or maintenance parts, tools, and accessories (see 3.27)
(s) Level of preservation and packaging and level of packing required (see 5.1 and 5.2).

6.3 Health precaution. The procuring agency will determine that the compound, when used for its intended purpose, is not likely to result in adverse effects on the health of personnel. Any questions raised regarding toxicity should be referred by the procuring agency to the departmental medical authority.

6.4 Electromagnetic compatibility. The contracting officer should submit a copy of the Interference Control Plan, the EMI/EMO Test Plan, and the Test Report required (see 4.6) to the Commanding General, U.S. Army Electronics Command, ATTN: AMSEL-RD-GF, Fort Monmouth, N.J. for approval.

6.5 Contract data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423 and invokes the provisions of paragraph 7-104.9(n) of the Defense Acquisition Regulations (DAR), the data requirements will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the Contract Data Requirements Lists (DD Form 1423) incorporated into the contract. When the provisions of DAR 7-104.9(n) are not invoked, the data shall be delivered in accordance with the contract requirements.

6.6 First article. When a first article is required, it shall be tested and approved under the appropriate provisions of paragraph 7-104.55 of the DAR. The first article should be a first production item consisting of one complete water softening unit or it may be a standard production item from the contractor's current inventory as specified in 4.3. The contracting officer should include specific intructions in all acquisition intruments, regarding arrangement for examinations, tests, and approval of the first article.

Custodians:

Army - ME
Navy - YD
Air Force - 99

Preparing activity:

Navy - YD

Project No. 4610-0078

User activity:

DLA - CS

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DOCUMENT IDENTIFIER *(Number)* AND TITLE
MIL-W-17121E - WATER SOFTENER UNIT, ZEOLITE PRESSURE TYPE

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

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